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Prof. Kenneth M. Merz, Jr., Editor-in-Chief
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Dear Prof. Merz,

We submit our manuscript **"Docking-Based Resistance-Aware Prioritization of Antimalarial Leads from African Natural Products with Targeted Molecular-Dynamics Follow-up"** for consideration as an Article in *JCIM.*

This work evaluates a docking-based resistance-aware prioritization strategy and uses targeted molecular dynamics as a separate structural follow-up; it does not assign a systems-level mechanism of action from docking alone. The 17 Set-C candidates were evaluated by docking-derived RRS across six resistance states, with ACSI and PNS providing complementary chemical-space and network-ranking dimensions; four separate parent-study complexes underwent targeted wild-type MD (10 ns each; 40 ns total). The target-balanced primary panel contained one Class A*, one Class A, four Class B, five Class C, and one Class D candidate; the five PfCRT-only records are reported separately as a coverage-limited sensitivity analysis. Only PfCRT–214 yielded an interpretable MM-GBSA estimate; the other parent-study systems were dissociated or conversion-excluded. Tartarus analysis provided no evidence of a composite MPO–affinity association, and this null result is interpreted as separation of ranking dimensions rather than proof of statistical independence.

A secondary 16-system Set-C MD pilot is reported only to quantify the extent to which trajectory-derived geometric and MM-GBSA summaries agree with docking-derived RRS. It is a single-replicate, 10-ns-per-system analysis of two candidates, is not merged into the primary docking-RRS estimand, and is not used as validation; mutant-to-wild-type ratios are interpreted strictly as retention within scoring noise rather than as gains.

Code, source data, and analysis records are available in the GitHub repository under an open-source licence. Statistical interpretation is bounded by effect sizes, seeded permutation p-values, bootstrap intervals, and the prespecified Bonferroni correction.

Suggested reviewers.

1. Dr. Fidele Ntie-Kang, University of Buea — African NP computational discovery
2. Dr. Kelly Chibale, University of Cape Town — antimalarial drug discovery
3. Dr. John D. Chodera, Memorial Sloan Kettering — free energy methods, MD best practices

This manuscript is original, not under consideration elsewhere, and all authors have approved the submission.

Sincerely,

Myke-Vital Temgoua Sao
(on behalf of all co-authors)